

Qatar-4b

R-2669

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_181026 · created 2026-08-30T01:03:51+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458417.69318 +/- 0.00062 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.038
Transit depth (Rp/Rs)^2	2.01 +/- 1.09 [%]
Orbital Inclination (inc)	89.89 +/- 1.85
Airmass coefficient 1 (a1)	1.715 +/- 0.059
Airmass coefficient 2 (a2)	-0.52 +/- 0.23
Scatter in the residuals of the lightcurve fit is	3.46 %
Best Comparison Star	None
Optimal Aperture	2.83
Optimal Annulus	4.76
Transit Duration (day)	0.0806 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.142 $\pm$ 0.038 vs 0.138 (deviation 0.09 σ)
Duration fit vs published	0.0806 d vs 0.0892 d (deviation 3.15 σ)
Mid-time O-C	1.2 min
Depth SNR	3.3
Residual scatter	3.46 %

Light curve

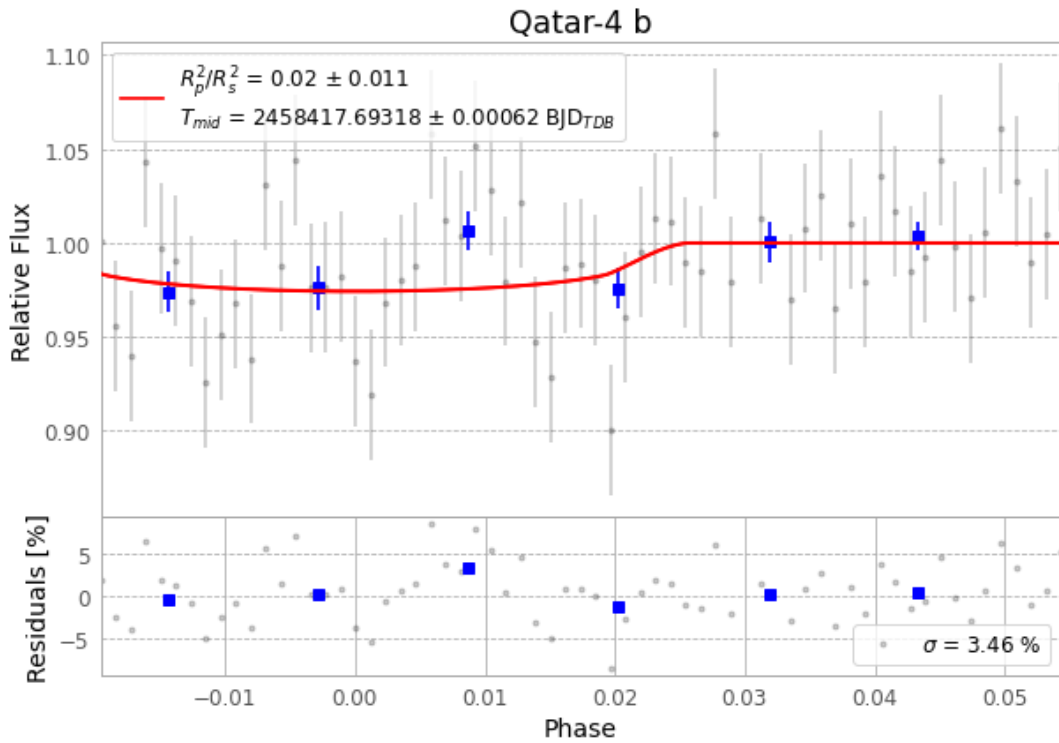


Figure 1 — FinalLightCurve_Qatar-4 b_2018-10-25.png (SHA-256 a196e77e20dad5d6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [394, 265], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9db133aaa891c17b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), Qatar-4181026033912.FITS → Qatar-4181026065112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-181026.log (a587963d8f56...), FinalLightCurve_Qatar-4 b_2018-10-25.png (a196e77e20da...), FinalLightCurve_Qatar-4 b_2018-10-25.csv (e963a245eb38...)

Compute resources

Wall time 66.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 01:03Z.